

VIT-IND-03

Smart Manufacturing: Acoustic AI-Based Metal Parts Quality Inspection & Predictive Maintenance

1. Project Objective

Thrust Area

Smart Manufacturing

Domain

Metal Parts & Industrial Quality Inspection

Core Problem

Manufacturing industries rely heavily on visual inspection and periodic maintenance. However:

- Internal cracks in metal components remain invisible
- Micro-fractures develop before failure
- Manual inspection is time-consuming
- Unexpected machine breakdowns cause production loss
- Traditional sensors may miss early-stage defects

Solution Concept

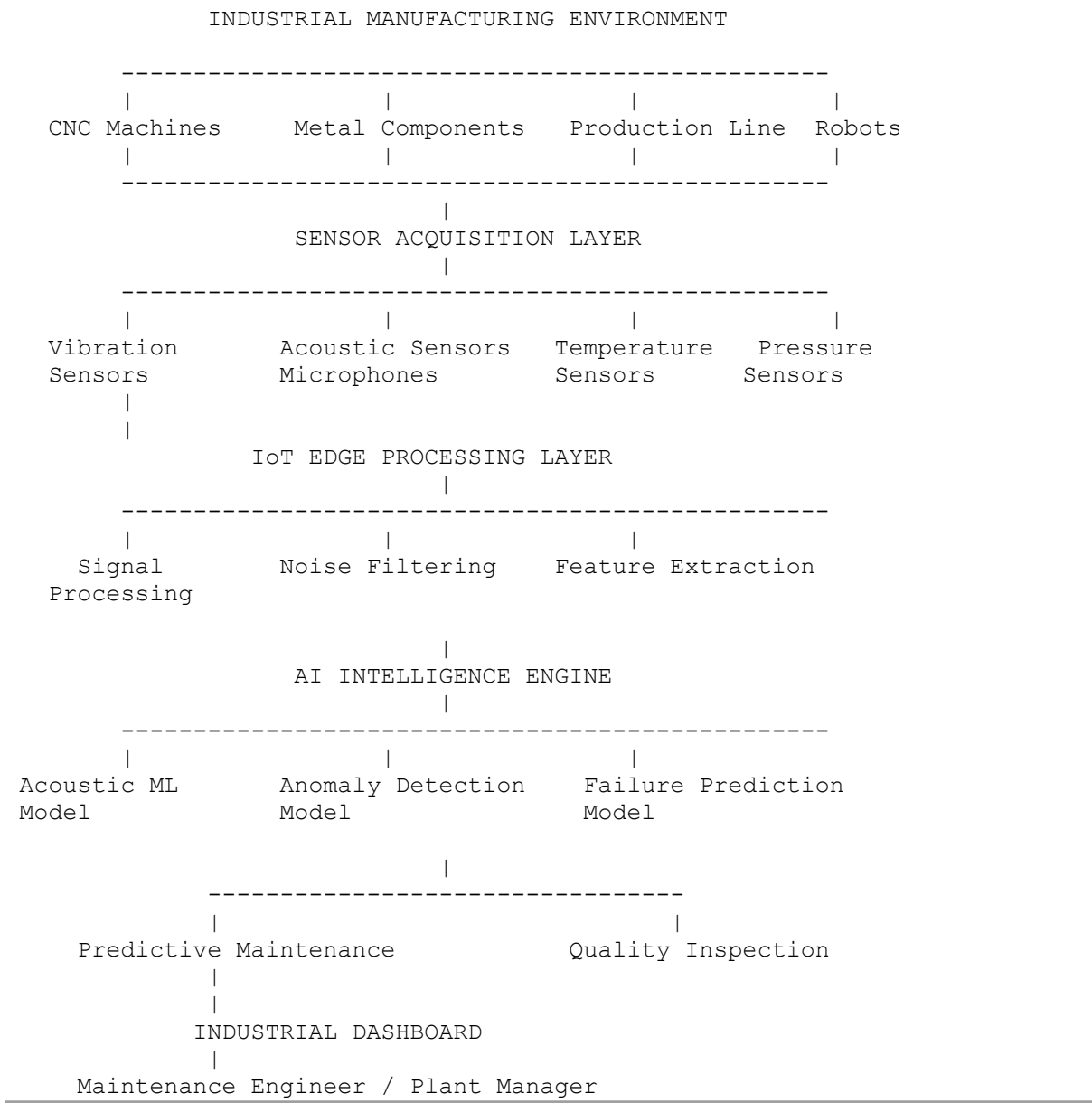
An **AI-powered Acoustic Predictive Quality Inspection Platform** that uses:

- Acoustic Machine Learning
- Signal Processing
- Vibration Analysis
- IoT Sensors
- Predictive Maintenance Models

The system listens to machine/component vibration and acoustic signatures to detect:

- Structural cracks
 - Material fatigue
 - Abnormal operating conditions
 - Failure probability
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2. High-Level System Architecture



3. Application Architecture Layers

Layer	Responsibility	Technology
Sensor Layer	Capture industrial signals	IoT Sensors, Microphones
Edge Layer	Real-time processing	Raspberry Pi, Edge AI
Data Layer	Store machine data	PostgreSQL, MongoDB
Signal Processing Layer	Convert raw signals	Librosa, SciPy, MATLAB
AI Layer	Defect detection & prediction	TensorFlow, PyTorch
Analytics Layer	Dashboard and reports	Power BI, React
Communication Layer	Device messaging	MQTT

Layer	Responsibility	Technology
Deployment Layer	Industrial scaling	Kubernetes

4. Database Architecture

Database modules:

1. Industrial Asset Management
2. Sensor Data Management
3. Acoustic Signal Processing
4. Defect Management
5. AI Model Management
6. Predictive Maintenance
7. Quality Analytics

A. INDUSTRIAL ASSET MASTER TABLES

1. TBL_MACHINE_MASTER

Purpose:
Stores all manufacturing equipment.

Field Name	Data Type	Description
Machine_ID	UUID	Primary Key
Machine_Name	VARCHAR	Equipment name
Machine_Type	VARCHAR	CNC, Lathe, Press
Manufacturer	VARCHAR	Machine company
Model_Number	VARCHAR	Machine model
Installation_Date	DATE	Installation
Production_Line_ID	FK	Manufacturing line
Criticality_Level	ENUM	High/Medium/Low
Machine_Status	ENUM	Active/Inactive

2. TBL_COMPONENT_MASTER

Purpose:
Stores metal components under inspection.

Field	Type	Description
Component_ID	UUID	Primary Key
Machine_ID	FK	Associated machine
Component_Name	VARCHAR	Part name
Material_Type	VARCHAR	Steel/Alloy etc.
Batch_Number	VARCHAR	Manufacturing batch
Manufacturing_Date	DATE	Production date
Expected_Life_Cycle	INTEGER	Usage hours

3. TBL_PRODUCTION_LINE_MASTER

Field	Type
Production_Line_ID	UUID
Line_Name	VARCHAR
Factory_Name	VARCHAR
Location	VARCHAR
Supervisor_ID	UUID
Status	VARCHAR

B. SENSOR DATA TABLES

4. TBL_SENSOR_MASTER

Purpose:
Stores IoT sensors.

Field	Type
Sensor_ID	UUID
Machine_ID	FK
Sensor_Type	VARCHAR
Manufacturer	VARCHAR
Installation_Date	DATE
Calibration_Date	DATE
Status	VARCHAR

Sensor Types:

- Acoustic Sensor
- Vibration Sensor
- Temperature Sensor
- Pressure Sensor

5. TBL_SENSOR_READING_LOG

Purpose:

Stores real-time sensor readings.

Field	Type
Reading_ID	UUID
Sensor_ID	FK
Timestamp	DATETIME
Signal_Value	FLOAT
Unit	VARCHAR
Sampling_Rate	FLOAT
Data_Quality	FLOAT

C. ACOUSTIC SIGNAL PROCESSING TABLES

6. TBL_AUDIO_SIGNAL_DATA

Purpose:

Stores raw acoustic recordings.

Field	Type
Audio_ID	UUID
Sensor_ID	FK
Recording_Time	DATETIME
File_Path	VARCHAR
Duration_Seconds	FLOAT
Frequency_Range	VARCHAR
Noise_Level	FLOAT

7. TBL_SIGNAL_FEATURE_STORE

Purpose:

Stores extracted ML features.

Field	Type
Feature_ID	UUID
Audio_ID	FK
Feature_Type	VARCHAR
Feature_Vector	JSON
Extraction_Method	VARCHAR
Created_Date	DATE

Examples:

- MFCC Features
- Spectral Energy
- Frequency Pattern
- Wavelet Features

D. DEFECT MANAGEMENT TABLES

8. TBL_DEFECT_CATEGORY_MASTER

Field	Type
Defect_ID	UUID
Defect_Name	VARCHAR
Description	TEXT
Severity_Level	ENUM

Examples:

- Crack
- Fatigue
- Corrosion
- Deformation

9. TBL_COMPONENT_INSPECTION_RESULT

Field	Type
Inspection_ID	UUID
Component_ID	FK
Model_ID	FK
Inspection_Date	DATE
Defect_Detected	BOOLEAN
Defect_Type	FK
Confidence_Score	FLOAT
Inspection_Status	VARCHAR

E. AI MODEL MANAGEMENT TABLES

10. TBL_AI_MODEL_MASTER

Purpose:
Stores acoustic AI models.

Field	Type
Model_ID	UUID
Model_Name	VARCHAR
Algorithm_Type	VARCHAR
Framework	VARCHAR
Version	VARCHAR
Deployment_Type	VARCHAR

Examples:

- CNN Acoustic Classifier
 - LSTM Failure Predictor
 - Anomaly Detection Model
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11. TBL_MODEL_TRAINING_HISTORY

Field	Type
Training_ID	UUID
Model_ID	FK
Dataset_Size	INTEGER
Training_Date	DATE
Accuracy	FLOAT
Precision	FLOAT

Field	Type
Recall	FLOAT
F1_Score	FLOAT

F. PREDICTIVE MAINTENANCE TABLES

12. TBL_MACHINE_HEALTH_SCORE

Purpose:
Continuous machine condition monitoring.

Field	Type
Health_ID	UUID
Machine_ID	FK
Health_Score	FLOAT
Remaining_Life_Percentage	FLOAT
Risk_Level	VARCHAR
Generated_Time	DATETIME

13. TBL_FAILURE_PREDICTION_RESULT

Field	Type
Prediction_ID	UUID
Machine_ID	FK
Model_ID	FK
Failure_Probability	FLOAT
Expected_Failure_Date	DATE
Recommended_Action	TEXT

G. MAINTENANCE MANAGEMENT TABLES

14.

TBL_MAINTENANCE_ACTION_MASTE R

Field	Type
Action_ID	UUID
Action_Name	VARCHAR
Description	TEXT
Estimated_Cost	FLOAT

Examples:

- Component Replacement
 - Lubrication
 - Calibration
 - Inspection
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15. TBL_MAINTENANCE_HISTORY

Field	Type
Maintenance_ID	UUID
Machine_ID	FK
Action_ID	FK
Technician_Name	VARCHAR
Start_Date	DATE
Completion_Date	DATE
Status	VARCHAR

H. INDUSTRIAL ANALYTICS TABLES

16. TBL_QUALITY_METRIC

Field	Type
Metric_ID	UUID
Machine_ID	FK
Production_Count	INTEGER
Defect_Count	INTEGER

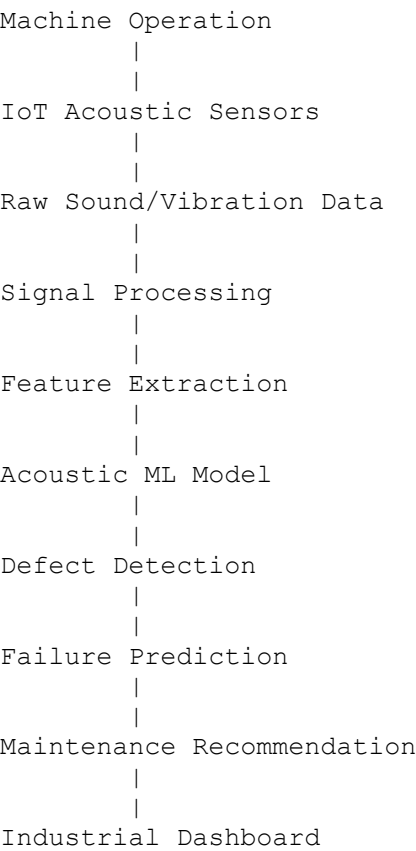
Field	Type
Quality_Rate	FLOAT
Generated_Date	DATE

17. TBL_ALERT_MASTER

Purpose:
Stores machine warnings.

Field	Type
Alert_ID	UUID
Machine_ID	FK
Alert_Type	VARCHAR
Severity	VARCHAR
Alert_Message	TEXT
Alert_Time	DATETIME
Status	VARCHAR

5. AI Processing Flow



6. AI Models Used

AI Component	Purpose
Acoustic ML Model	Detect abnormal sound patterns
CNN Model	Classify defect signatures
LSTM Model	Predict future failure
Anomaly Detection	Detect unknown defects
Predictive Analytics	Estimate component life

7. Expected Platform Features

Feature ID	Feature
F01	Real-time machine health monitoring
F02	Acoustic defect detection
F03	Crack prediction engine
F04	Predictive maintenance scheduler
F05	Digital maintenance history
F06	Production quality analytics
F07	AI-based failure alerts

8. Expected Outcomes

Outcome	Measurement KPI
Reduced machine downtime	Downtime hours reduced
Early defect detection	Detection accuracy
Improved product quality	Defect percentage reduction
Lower maintenance cost	Cost savings
Extended equipment life	Remaining useful life improvement

9. Industrial Impact

Sector	Impact
Automotive	Metal component quality assurance
Aerospace	Safety-critical inspection
Heavy Engineering	Machine failure prevention
Manufacturing Plants	Industry 4.0 adoption

10. SDG Alignment

SDG	Contribution
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SDG 9	Smart industrial infrastructure
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SDG 12	Efficient resource utilization
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This architecture converts **VIT-IND-03** into a complete **Industry 4.0 Acoustic Intelligence Platform** with database-ready design suitable for implementation in PROSPERO MIS / Smart Manufacturing module.